ASSIGNEMENT-1 ON MAT 3141 (Applied Statistics & Probability) using Python

```
import pandas as pd
import numpy as np
from scipy import stats
import matplotlib.pyplot as plt
import seaborn as sns
```

1. Student Data (Group of 5 students - CSE 5th Semester)

```
data = {
    'ID': ['241311049', '241311050', '241311051', '241311052', '241311053'],
    'Name': ['MST. SADIA AKTER MUNIA', 'MST. SONIYA AKTER', 'ABDULLAH ATIF', 'HABIBATU
    'Gender': ['F', 'F', 'M', 'F', 'M'],
    'Home District': ['Rajshahi', 'Rajshahi', 'Rajshahi', 'Chapainawabganj', 'Natore']
    'Age': [22, 21, 21, 21, 22],
    'Parents Income': [55000, 30000, 60000, 10000, 22000],   # in BDT
    'CGPA': [3.18, 3.60, 3.46, 2.79, 3.25]
}

df = pd.DataFrame(data)

print("1.b) Individual Results:")
print(df.to_string(index=False))

print("\n1.c) Combined Result (DataFrame):")
print(df)
```

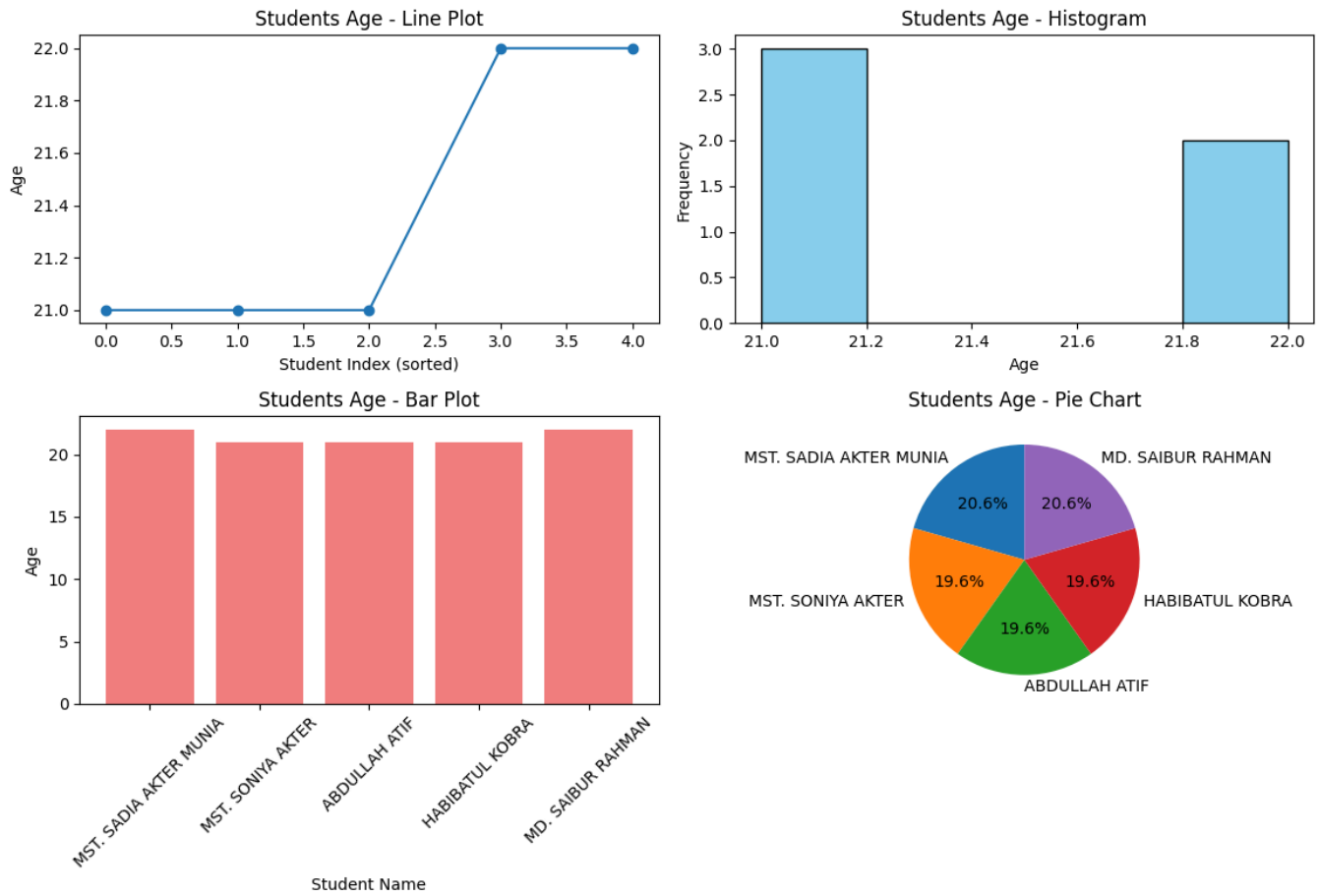
```
1.b) Individual Results:
       ID                   Name Gender   Home District  Age  Parents Income  CGPA
241311049 MST. SADIA AKTER MUNIA      F        Rajshahi   22           55000  3.18
241311050      MST. SONIYA AKTER      F        Rajshahi   21           30000  3.60
241311051          ABDULLAH ATIF      M        Rajshahi   21           60000  3.46
241311052        HABIBATUL KOBRA      F Chapainawabganj   21           10000  2.79
241311053      MD. SAIBUR RAHMAN      M          Natore   22           22000  3.25

1.c) Combined Result (DataFrame):
          ID                   Name Gender    Home District  Age  \
0  241311049  MST. SADIA AKTER MUNIA      F         Rajshahi   22
1  241311050       MST. SONIYA AKTER      F         Rajshahi   21
2  241311051           ABDULLAH ATIF      M         Rajshahi   21
3  241311052         HABIBATUL KOBRA      F  Chapainawabganj   21
4  241311053       MD. SAIBUR RAHMAN      M           Natore   22

   Parents Income  CGPA
0           55000  3.18
1           30000  3.60
2           60000  3.46
3           10000  2.79
4           22000  3.25
```

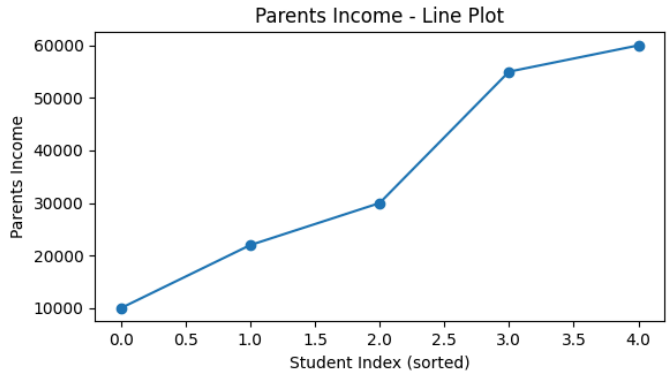
```python
print("\n1.d) Measures of Central Tendency:")

for col in ['Age', 'Parents Income', 'CGPA']:
    col_data = df[col]
    print(f"\n--- {col} ---")
    print(f"Arithmetic Mean (AM)  : {np.mean(col_data):.2f}")
    print(f"Geometric Mean (GM)   : {stats.gmean(col_data):.2f}")
    print(f"Harmonic Mean (HM)    : {stats.hmean(col_data):.2f}")
    print(f"Median                : {np.median(col_data):.2f}")
    mode_result = stats.mode(col_data, keepdims=True)
    print(
        f"Mode                  : {mode_result.mode[0]} (count: {mode_result.count[0]}
```

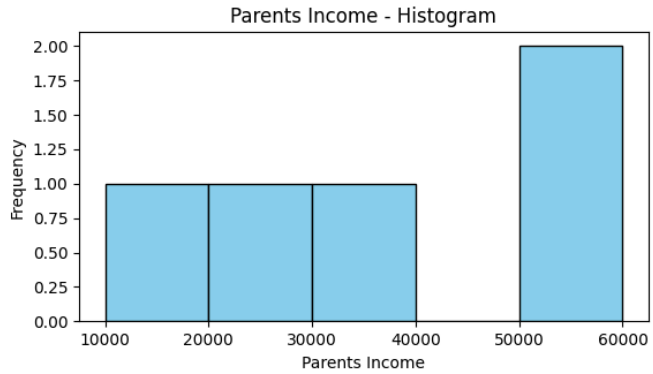
```
1.d) Measures of Central Tendency:

--- Age ---
Arithmetic Mean (AM)  : 21.40
Geometric Mean (GM)   : 21.39
Harmonic Mean (HM)    : 21.39
Median                : 21.00
Mode                  : 21 (count: 3)

--- Parents Income ---
Arithmetic Mean (AM)  : 35400.00
Geometric Mean (GM)   : 29350.23
Harmonic Mean (HM)    : 23404.26
Median                : 30000.00
Mode                  : 10000 (count: 1)

--- CGPA ---
Arithmetic Mean (AM)  : 3.26
Geometric Mean (GM)   : 3.24
Harmonic Mean (HM)    : 3.23
Median                : 3.25
Mode                  : 2.79 (count: 1)
```

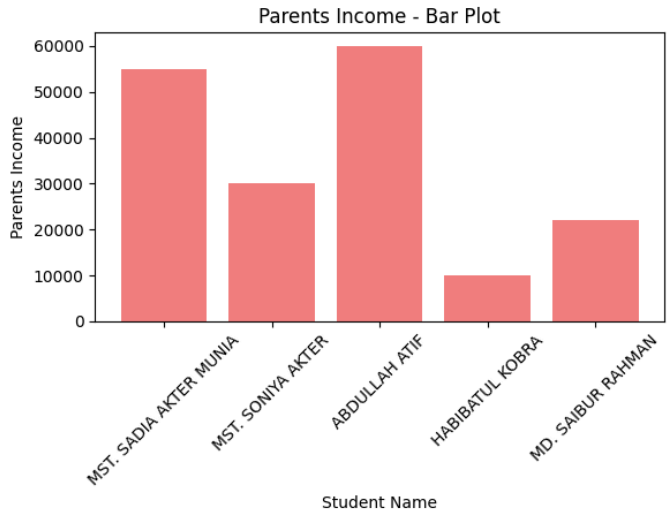
```python
# Function to draw all 4 plots for a column
def draw_plots(column_name, data_series, title_prefix):
    plt.figure(figsize=(12, 8))

    # Line plot (sorted values)
    plt.subplot(2, 2, 1)
    plt.plot(sorted(data_series), marker='o')
    plt.title(f'{title_prefix} - Line Plot')
    plt.xlabel('Student Index (sorted)')
    plt.ylabel(column_name)

    # Histogram
    plt.subplot(2, 2, 2)
    plt.hist(data_series, bins=5, color='skyblue', edgecolor='black')
    plt.title(f'{title_prefix} - Histogram')
    plt.xlabel(column_name)
    plt.ylabel('Frequency')

    # Bar plot (by Name)
    plt.subplot(2, 2, 3)
```

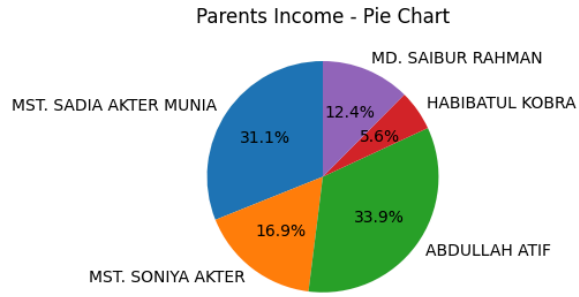
```
    plt.bar(df['Name'], data_series, color='lightcoral')
    plt.title(f'{title_prefix} - Bar Plot')
    plt.xlabel('Student Name')
    plt.ylabel(column_name)
    plt.xticks(rotation=45)

    # Pie chart (by Name - for demonstration)
    plt.subplot(2, 2, 4)
    plt.pie(data_series, labels=df['Name'], autopct='%1.1f%%', startangle=90)
    plt.title(f'{title_prefix} - Pie Chart')

    plt.tight_layout()
    plt.show()


# Draw for all three variables
draw_plots('Age', df['Age'], 'Students Age')
draw_plots('Parents Income', df['Parents Income'], 'Parents Income')
draw_plots('CGPA', df['CGPA'], 'CGPA')
```

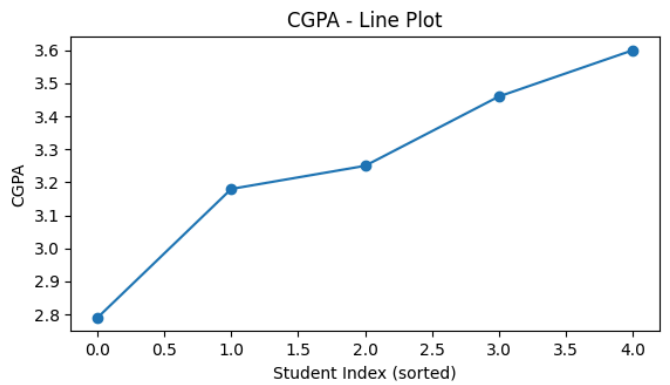


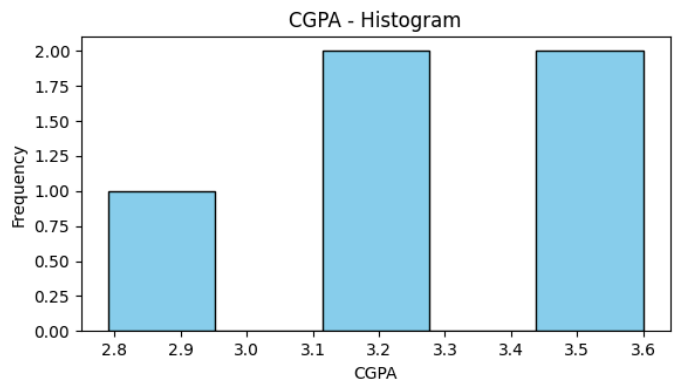

Parents Income - Line Plot
Parents Income
Student Index (sorted)

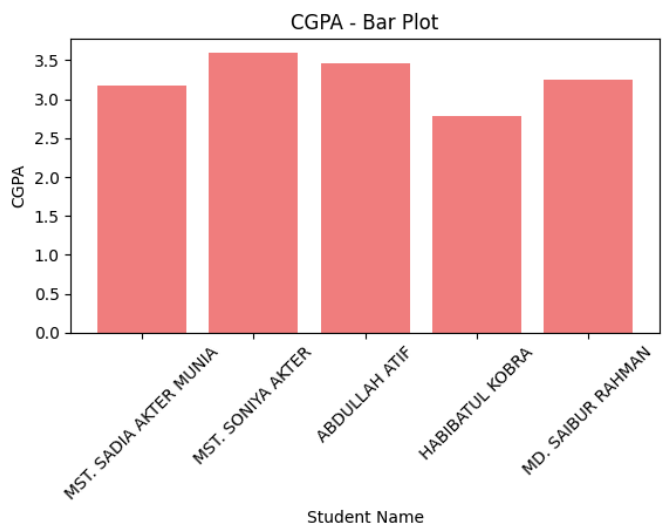

Parents Income - Histogram
Frequency
Parents Income

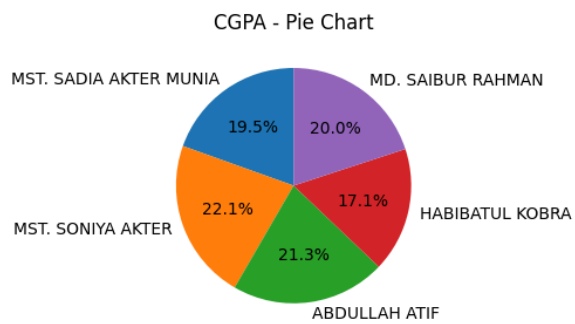

Parents Income - Bar Plot
Parents Income
MST. SADIA AKTER MUNIA
MST. SONIYA AKTER
ABDULLAH ATIF
HABIBATUL KOBRA
MD. SAIBUR RAHMAN
Student Name


Parents Income - Pie Chart
MD. SAIBUR RAHMAN
MST. SADIA AKTER MUNIA
HABIBATUL KOBRA
12.4%
5.6%
31.1%
33.9%
16.9%
ABDULLAH ATIF
MST. SONIYA AKTER

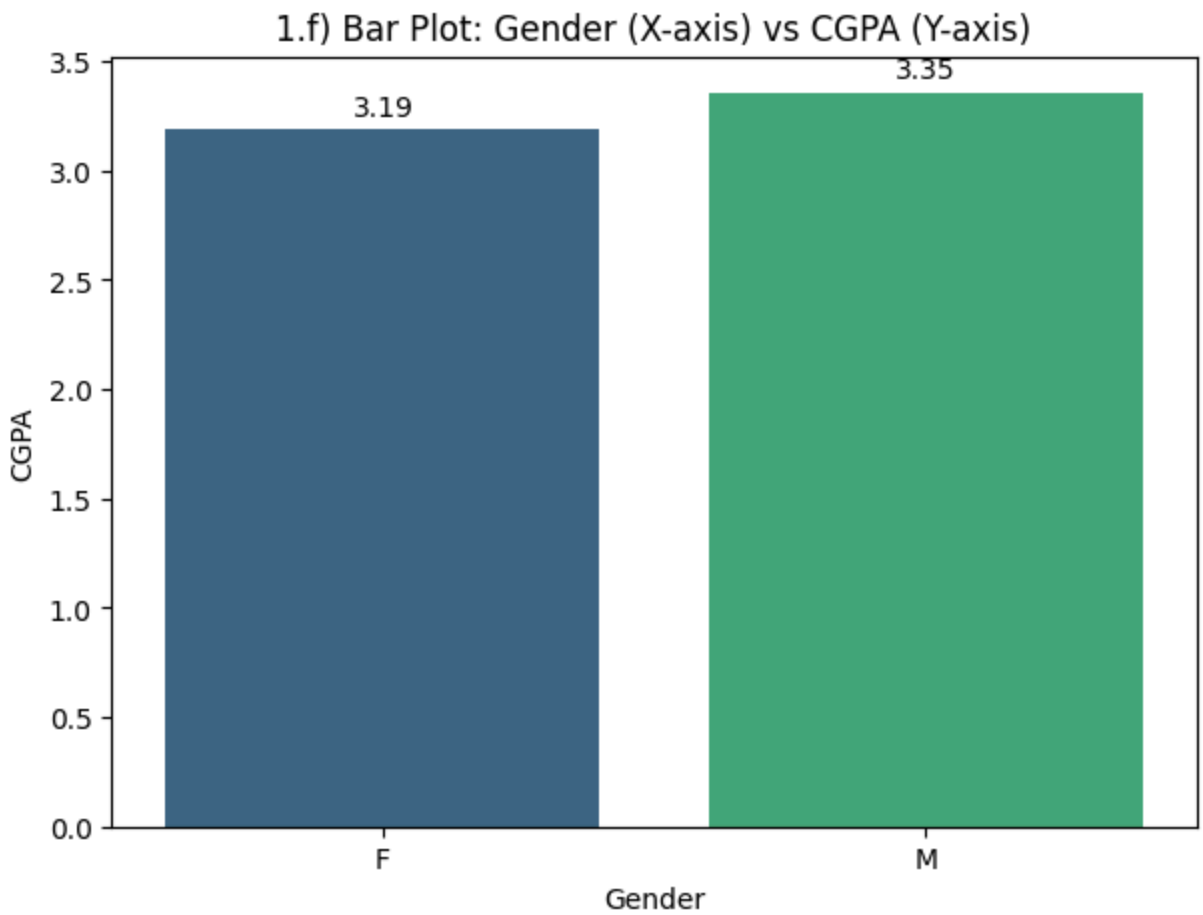

CGPA - Line Plot
CGPA
Student Index (sorted)


CGPA - Histogram
Frequency
CGPA


CGPA - Bar Plot
CGPA
MST. SADIA AKTER MUNIA
MST. SONIYA AKTER
ABDULLAH ATIF
HABIBATUL KOBRA
MD. SAIBUR RAHMAN
Student Name


CGPA - Pie Chart
MST. SADIA AKTER MUNIA
MD. SAIBUR RAHMAN
19.5%
20.0%
22.1%
17.1%
HABIBATUL KOBRA
MST. SONIYA AKTER
21.3%
ABDULLAH ATIF

```
plt.figure(figsize=(7, 5))

ax = sns.barplot(x='Gender', y='CGPA', data=df, hue='Gender',
                 palette='viridis',  errorbar=None, legend=False)

# Add CGPA values on top of bars
for container in ax.containers:
    ax.bar_label(container, fmt='%.2f', padding=3)

plt.title('1.f) Bar Plot: Gender (X-axis) vs CGPA (Y-axis)')
plt.xlabel('Gender')
plt.ylabel('CGPA')
plt.show()
```



2. Statistical Analysis & Visualization of Variable X

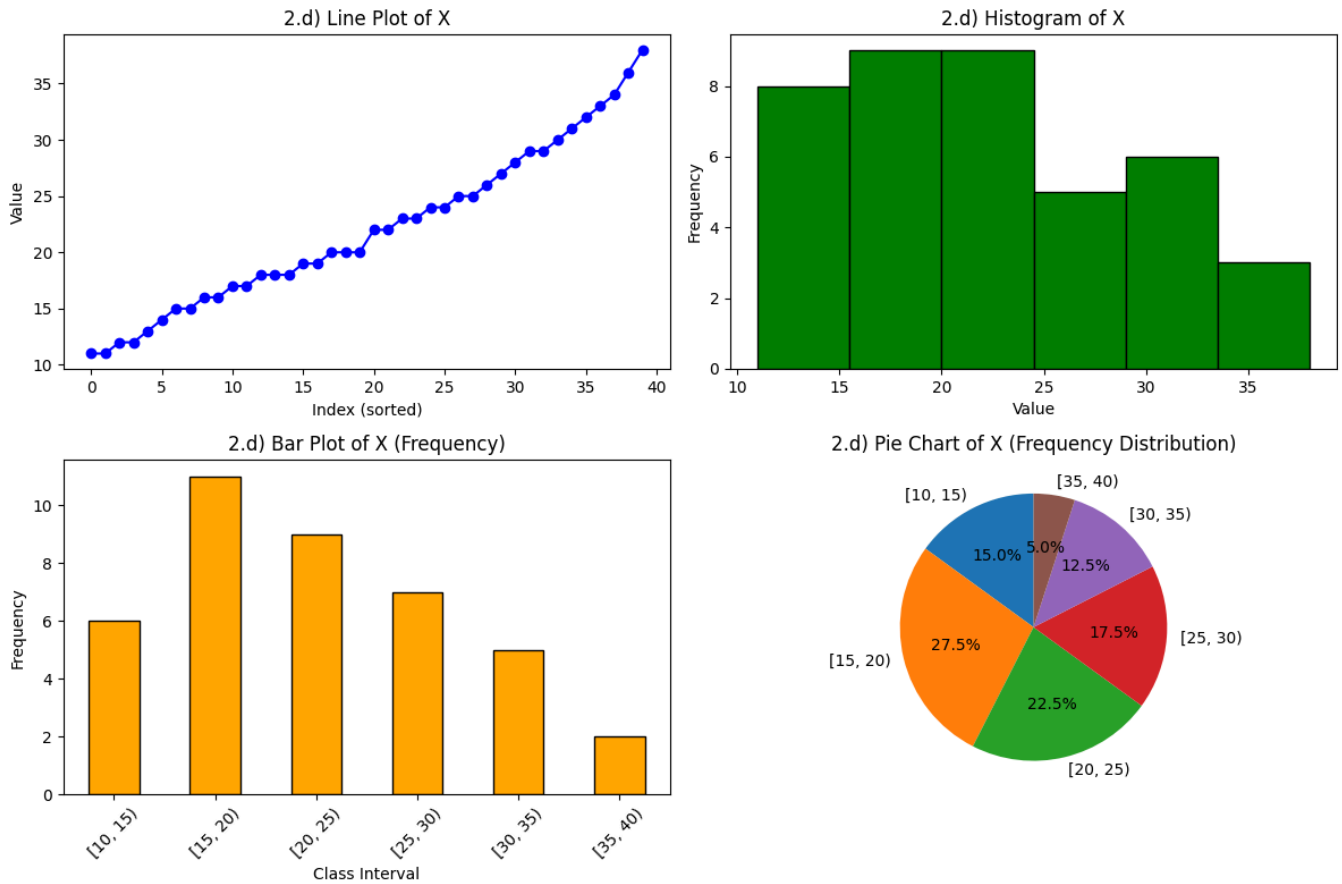
```
# 2.a) Input the data
X = np.array([11, 17, 15, 22, 11, 16, 19, 24, 29, 25,
              18, 18, 25, 26, 32, 14, 17, 20, 23, 27,
              30, 12, 24, 36, 18, 15, 20, 28, 33, 38,
              34, 13, 12, 16, 20, 22, 29, 19, 23, 31])

print("2.a) Data loaded successfully (40 values)")

# 2.b) AM, GM, HM, Median, Mode
print("\n2.b) Measures of Central Tendency for X:")
```

```python
print(f"Arithmetic Mean (AM)  : {np.mean(X):.2f}")
print(f"Geometric Mean (GM)   : {stats.gmean(X):.2f}")
print(f"Harmonic Mean (HM)    : {stats.hmean(X):.2f}")
print(f"Median                : {np.median(X):.2f}")
mode_result = stats.mode(X, keepdims=True)
print(
    f"Mode                  : {mode_result.mode[0]} (count: {mode_result.count[0]})")

# 2.c) Frequency Distribution Table (class width = 5, suitable for range 11-38)
bins = np.arange(10, 41, 5)
freq_table = pd.cut(X, bins=bins, right=False).value_counts().sort_index()
print("\n2.c) Frequency Distribution Table (Class Interval = 5):")
print(freq_table)
```

```
2.a) Data loaded successfully (40 values)

2.b) Measures of Central Tendency for X:
Arithmetic Mean (AM)  : 22.05
Geometric Mean (GM)   : 20.88
Harmonic Mean (HM)    : 19.74
Median                : 21.00
Mode                  : 18 (count: 3)

2.c) Frequency Distribution Table (Class Interval = 5):
[10, 15)     6
[15, 20)    11
[20, 25)     9
[25, 30)     7
[30, 35)     5
[35, 40)     2
Name: count, dtype: int64
```

```python
plt.figure(figsize=(12, 8))

# Line plot (sorted)
plt.subplot(2, 2, 1)
plt.plot(np.sort(X), marker='o', color='blue')
plt.title('2.d) Line Plot of X')
plt.xlabel('Index (sorted)')
plt.ylabel('Value')

# Histogram
plt.subplot(2, 2, 2)
plt.hist(X, bins=6, color='green', edgecolor='black')
plt.title('2.d) Histogram of X')
plt.xlabel('Value')
plt.ylabel('Frequency')

# Bar plot (using frequency table)
plt.subplot(2, 2, 3)
freq_table.plot(kind='bar', color='orange', edgecolor='black')
plt.title('2.d) Bar Plot of X (Frequency)')
plt.xlabel('Class Interval')
plt.ylabel('Frequency')
plt.xticks(rotation=45)
```

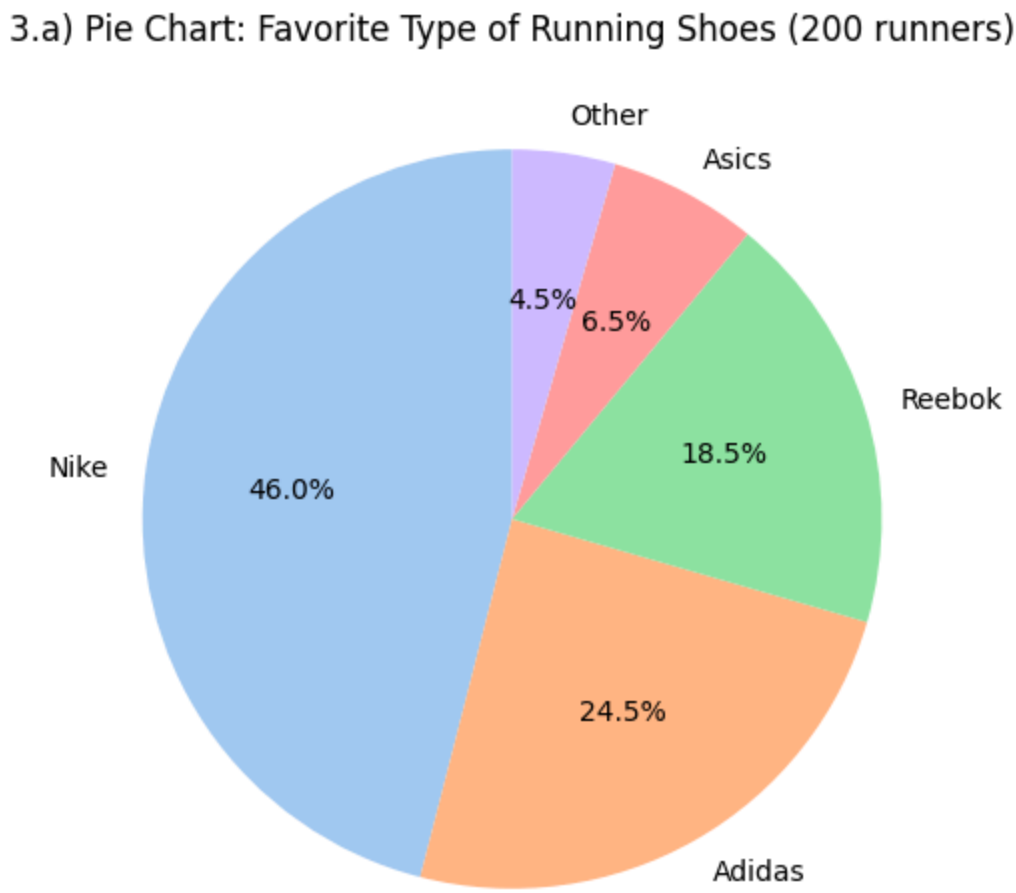
```
# Pie chart (using frequency table)
plt.subplot(2, 2, 4)
plt.pie(freq_table.values, labels=freq_table.index.astype(
    str), autopct='%1.1f%%', startangle=90)
plt.title('2.d) Pie Chart of X (Frequency Distribution)')

plt.tight_layout()
plt.show()
```

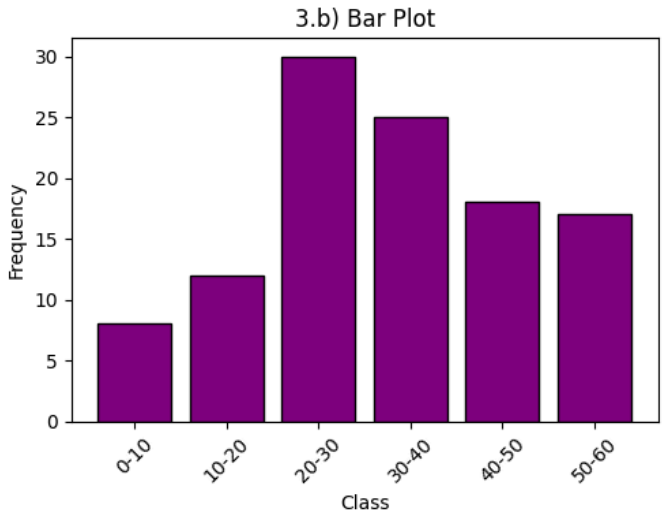


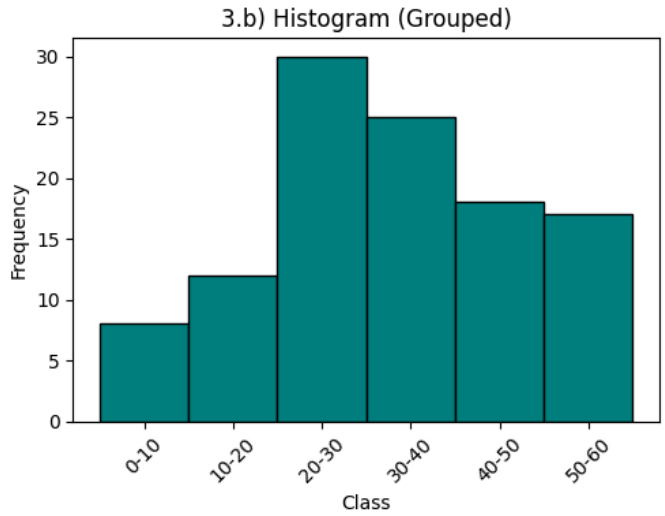
3. Pie Chart, Histogram and Bar Plot of Categorical & Grouped Data

```
# 3.a) Pie chart for favorite running shoes
shoes = ['Nike', 'Adidas', 'Reebok', 'Asics', 'Other']
runners = [92, 49, 37, 13, 9]

plt.figure(figsize=(6, 6))
plt.pie(runners, labels=shoes, autopct='%1.1f%%',
        startangle=90, colors=sns.color_palette('pastel'))
plt.title('3.a) Pie Chart: Favorite Type of Running Shoes (200 runners)')
plt.show()

# 3.b) Histogram & Bar plot for given frequency table
classes = ['0-10', '10-20', '20-30', '30-40', '40-50', '50-60']
freq = [8, 12, 30, 25, 18, 17]

plt.figure(figsize=(10, 4))

# Bar Plot
```

```python
plt.subplot(1, 2, 1)
plt.bar(classes, freq, color='purple', edgecolor='black')
plt.title('3.b) Bar Plot')
plt.xlabel('Class')
plt.ylabel('Frequency')
plt.xticks(rotation=45)

# Histogram (same data, grouped style)
plt.subplot(1, 2, 2)
plt.bar(classes, freq, width=1.0, align='center',
        color='teal', edgecolor='black')
plt.title('3.b) Histogram (Grouped)')
plt.xlabel('Class')
plt.ylabel('Frequency')
plt.xticks(rotation=45)

plt.tight_layout()
plt.show()
```




3.b) Bar Plot
Frequency
Class
0
5
10
15
20
25
30
0-10
10-20
20-30
30-40
40-50
50-60


3.b) Histogram (Grouped)
Frequency
Class
0
5
10
15
20
25
30
0-10
10-20
20-30
30-40
40-50
50-60